SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Pradeepa S	**Reg. No.:**	192424318
Date:	12/10/2026	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>*.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Problem Statement

Online Course Learning Management System: Develop an e-learning platform where instructors can upload courses and learning materials while students can enroll, attend assessments, track progress, and receive certificates.

1. Project Vision and Stakeholders

Project Vision

The vision of the **Online Course Learning Management System (LMS)** is to develop a user-friendly e-learning platform that enables instructors to create and upload courses and learning materials, while allowing students to enroll in courses, attend assessments, monitor their learning progress, and receive certificates after successful completion.

Objectives

- To provide a platform for instructors to upload and manage courses.
- To allow students to easily enroll in available courses.
- To provide online assessments for students.
- To track student learning progress.
- To generate certificates for successfully completed courses.
- To maintain course and student information in a centralized system.

Stakeholders

Stakeholder	Role
Students	Enroll in courses, access materials, attend assessments, track progress and receive certificates
Instructors	Create courses, upload learning materials and conduct assessments
Administrator	Manage users, courses and overall system
Institution/Management	Monitor the overall learning platform and its performance

2. Epics and User Stories

Epic 1 – User Management

User Story 1:
As a student, I want to create and log in to my account, so that I can access the learning platform.

User Story 2:
As an administrator, I want to manage student and instructor accounts, so that user information is maintained properly.

Epic 2 – Course Management

User Story 3:
As an instructor, I want to create and upload courses, so that students can access the courses online.

User Story 4:
As an instructor, I want to upload learning materials, so that students can study the course content.

Epic 3 – Course Enrollment

User Story 5:
As a student, I want to view available courses and enroll in a course, so that I can start learning.

Epic 4 – Assessment

User Story 6:
As an instructor, I want to create assessments, so that I can evaluate student learning.

User Story 7:
As a student, I want to attend online assessments, so that I can test my knowledge.

Epic 5 – Progress Tracking

User Story 8:
As a student, I want to track my course progress, so that I can know how much of the course I have completed.

Epic 6 – Certificate Management

User Story 9:
As a student, I want to receive a certificate after successfully completing a course, so that I can prove my achievement.

3. Acceptance Criteria

The assessment requires acceptance criteria to be **clear, measurable and testable**, preferably using Given–When–Then format. CSA10-CO1 AT2 Agile Sprint Plan…

User Story 1 – Student Login

- **Given** the student has a registered account,
- **When** the student enters valid login credentials,
- **Then** the system should allow the student to access the LMS.

User Story 3 – Upload Course

- **Given** the instructor is logged in,
- **When** the instructor enters course details and uploads the course,
- **Then** the system should save and display the course.

User Story 4 – Upload Learning Materials

- **Given** an instructor has created a course,
- **When** the instructor uploads learning materials,
- **Then** the materials should be available to enrolled students.

User Story 5 – Course Enrollment

- **Given** the student is logged in,
- **When** the student selects an available course and clicks enroll,
- **Then** the student should be enrolled in that course.

User Story 7 – Attend Assessment

- **Given** the student is enrolled in a course,
- **When** the student starts an available assessment and submits the answers,
- **Then** the system should record the assessment result.

User Story 8 – Track Progress

- **Given** the student is enrolled in a course,
- **When** the student completes learning activities,
- **Then** the system should update and display the student's progress.

User Story 9 – Certificate

- **Given** the student has successfully completed the course,
- **When** the completion requirements are satisfied,
- **Then** the system should generate and provide a certificate.

4. Product Backlog Prioritization

The backlog is prioritized according to **business value, customer needs and dependencies**, as required by the assessment.

Priority	User Story	Story Points
1-High	Student registration and login	3
2-High	Instructor creates and uploads courses	5
3-High	Instructor uploads learning materials	5
4-High	Student views and enrolls in courses	5
5-High	Student accesses course materials	3
6-Medium	Instructor creates assessments	5
7-Medium	Student attends assessments	5
8-Medium	Student tracks progress	5
9-Low	Certificate generation	3
10-Low	Administrator management	5

5. Sprint Backlog – Sprint 1

The first sprint should focus on the **core LMS functionality** needed to establish the platform.

Sprint 1 Selected User Stories

1. Student registration and login
2. Instructor creates and uploads courses
3. Instructor uploads learning materials
4. Student views available courses
5. Student enrolls in a course

Sprint Tasks

Task 1 – User Authentication

- Create student/instructor registration.
- Create login functionality.
- Validate user credentials.

Task 2 – Course Management

- Create course form.
- Add course details.
- Implement course upload and storage.

Task 3 – Learning Materials

- Create material upload functionality.
- Link materials with courses.
- Allow enrolled students to access materials.

Task 4 – Course Enrollment

- Display available courses.
- Add enrollment option.
- Store enrollment information.

6. Story Point Estimation

The **Fibonacci sequence (1, 2, 3, 5, 8...)** is used for relative effort estimation, as suggested in the assessment.

User Story	Story Points	Justification
Student registration and login	3	Simple authentication functionality
Instructor uploads courses	5	Requires course form, validation and storage
Instructor uploads materials	5	Requires file handling and course linking
Student views courses	3	Basic course listing functionality
Student enrolls in course	5	Requires enrollment logic and database update

7. Sprint Goal and Expected Deliverables

Sprint Goal

To develop the basic functional foundation of the Online Course Learning Management System by implementing user authentication, course creation, learning material upload, course listing and student enrollment.

Expected Deliverables

At the end of Sprint 1:

- Student and instructor registration/login should work.
- Instructors should be able to create and upload courses.
- Instructors should be able to upload learning materials.
- Students should be able to view available courses.
- Students should be able to enroll in courses.

- Basic course and enrollment information should be stored in the system.

Conclusion

The Agile Sprint Planning approach helps the development team divide the Online Course Learning Management System into manageable **epics, user stories and sprint tasks**. By prioritizing high-value features and estimating them using story points, the team can develop the LMS incrementally and deliver useful functionality within each sprint.